

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 1

Year: 2019

Subject: 353152 Electronic Circuits I

Section: 5-6

Date: 16 August 2019

Time: 15.00-17.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. The test is divided into 2 sections. There will be 13 pages (including this page) and they are worth 89 points.

- Answer the questions on this test papers.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators can be used in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

Part 1 (64 points) Multiple choices

Instruction: Mark the correct answer for the following questions in the provided answer sheet.

1. Valence shell of good conductor contain _____ electrons.
 (a) 1 (b) 4 (c) 6 (d) 8

2. Which type of material has widest energy gap between conduction band and valence band?
 (a) insulator (b) semiconductor
 (c) conductor (d) resistor

3. Atoms of semiconductors contain _____ valence electrons.
 (a) 1 (b) 4 (c) 6 (d) 8

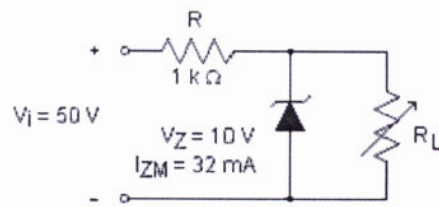
4. How much voltage drop over silicon diode when it is in forward bias mode?
 (a) 0.2 V (b) 0.3 V (c) 0.6 V (d) 0.7 V

5. The voltage drops of a Germanium diode are _____ V when they are operating in forward bias mode.
 (a) 0.2 V (b) 0.3 V (c) 0.6 V (d) 0.7 V

6. Which circuit provides output as $V_{DC} = 0.318V_m$ for ideal case?
 (a) Half Wave Rectifier (b) Bridge Rectifier
 (c) Full-Wave Rectifier (d) Center-Tapped Transformer Rectifier

7. Which one of the following is true about Half-wave rectifier?
 (a) Half-wave rectifier has lower voltage drop at diode compare with bridge-rectifier.
 (b) The cost of bridge-rectifier is lower than Half-wave rectifier.
 (c) Half-wave rectifier provides higher dc voltage output than full-wave rectifier.
 (d) Half-wave rectifier contains more number of diodes over center-tapped transformer rectifier.

8. Which of the following transistor operating mode represents the open circuit?
(a) Cut-off (b) Saturation (c) Active (d) None
9. Which of the following is true about Cutoff-region?
(a) EB Junction is forward biased.
(b) EB Junction is reverse biased.
(c) CB Junction is forward biased.
(d) CE Junction is reverse biased.
10. Which diode(s) has (have) a zero level current and voltage drop in the ideal model?
(a) Si (b) Ge (c) Both Si and Ge (d) Neither Si nor Ge
11. An open circuit can have any voltage across its terminals, but the current is always _____.
(a) 5 A (b) 0 A (c) 1 A (d) ∞
12. The intersection of the load line with the characteristic curve determines the _____ of the system.
(a) point of operation (b) load-line analysis
(c) characteristic curve (d) forward bias
13. The quiescent point (Q-point) is defined by a(n) _____.
(a) ac network (b) dc network
(c) ac and dc network (d) KVL
14. A germanium diode is approximated by _____ equivalent for voltages less than 0.3 V.
(a) a short circuit (b) series circuit
(c) a parallel circuit (d) an open circuit
15. With this Zener diode in its "on state," what is the level of I_Z (current of Zener diode) for the maximum load resistance?

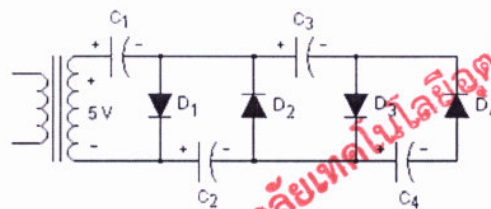


- (a) 0 mA (b) $I_Z = 8\text{mA}$ (c) $I_Z = I_{RL} = 40\text{mA}$ (d) $I_Z = 32\text{mA}$

16. What is the peak inverse voltage across each diode in a voltage doubler?

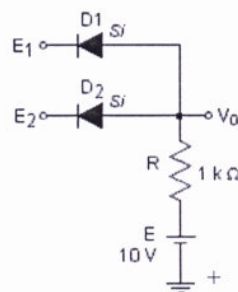
- (a) V_m (b) $2V_m$ (c) $0.5V_m$ (d) $0.25V_m$

17. What is the voltage measured from the negative terminal of C_4 to the negative terminal of the transformer?



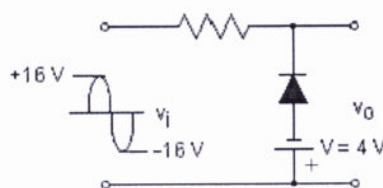
- (a) -10 V (b) -20 V (c) 10 V (d) 20 V

18. Determine the current through each diode if $E_1 = E_2 = 0\text{ V}$.



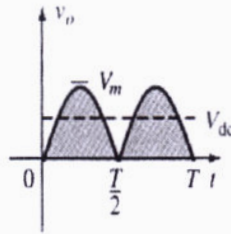
- (a) 4.65 mA (b) 9.3 mA (c) 18.68 mA (d) 0.7 mA

19. Determine the peak for both half cycles of the output waveform. Assume that the diode is ideal.



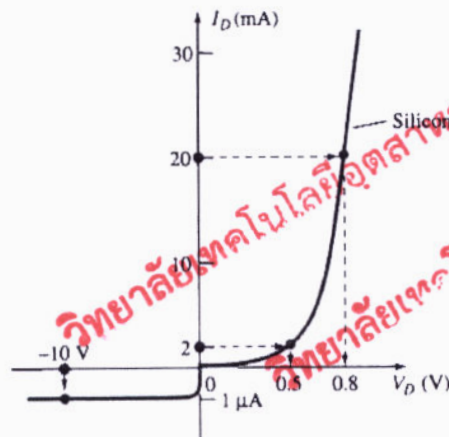
- (a) 16 V, -4 V (b) 16 V, 4 V (c) -16 V, 4 V (d) -16 V, -4 V

20. Which of the following is the average output value of full-wave rectifier?



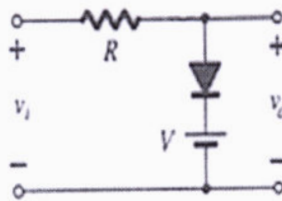
- (a) $0.318V_m$ (b) $0.5V_m$ (c) $0.636V_m$ (d) $1.414V_m$

21. From figure below, what is the DC resistance at 20 mA?



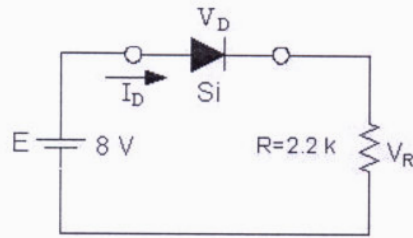
- (a) $4 \text{ m}\Omega$ (b) 4Ω (c) 40Ω (d) $40 \text{ k}\Omega$

22. What is the output voltage of circuit below? Assume that the diodes are ideal and sine wave input.



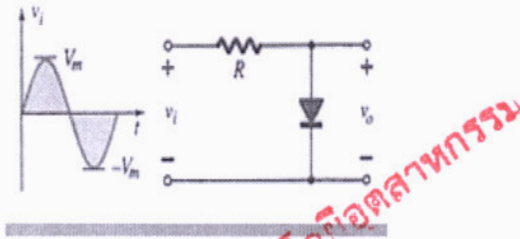
- (a) (b) (c) (d)

23. Find V_D and V_R



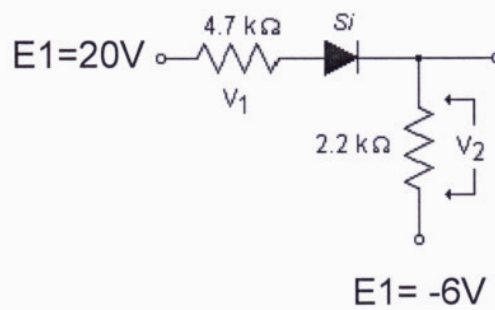
- (a) 0.7V and 7.3V (b) 0.3V and 7.7V (c) 0.7V and 0.3V (d) 0.3 V and 8V

24. What is the output voltage of the ideal diode circuit as shown in figure below?



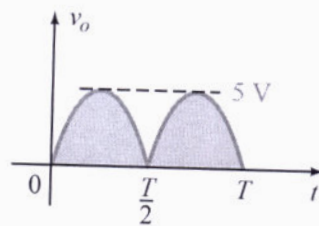
- (a) (b) (c) (d)

25. Determine V_2 .



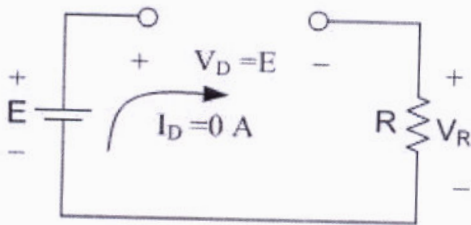
- (a) 3.201 V (b) 0 V (c) 4.3 V (d) 1.371 V

26. Which of the following is the V_{dc} of the waveform obtained from the silicon bridge rectifier?



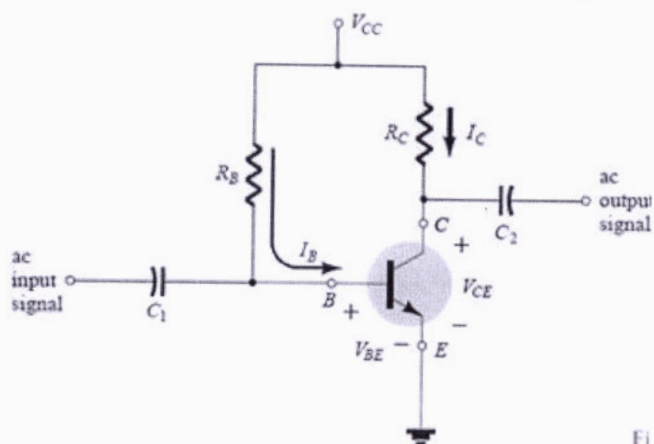
- (a) 3.18 V (b) 5 V (c) 2.29 V (d) 1.59 V

27. Find V_R where the given $E=20V$.



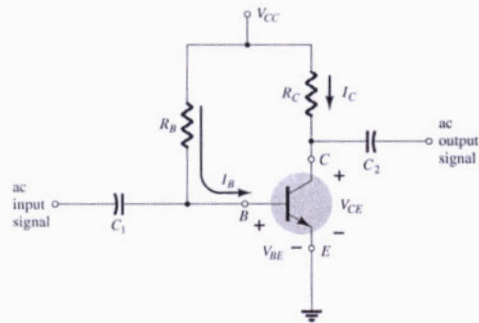
- (a) 0.7V
(b) 0.3V
(c) 0 V
(d) 20V

28. What is the purpose of capacitor C_1 in circuit below.



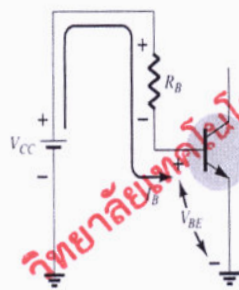
- (a) DC block (b) voltage multiplier (c) Filter (d) Clipper

29. What is a bias type of the circuit in figure below?



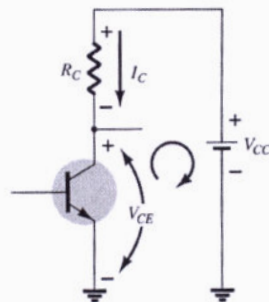
- (a) Fixed-bias circuit
(b) Emitter-stabilized bias circuit
(c) Voltage divider bias circuit
(d) DC bias with voltage feedback

30. From Base-Emitter loop in picture below, what is the Kirchhoff's voltage equation for the loop?



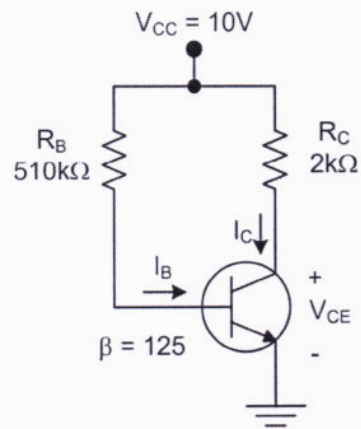
- (a) $V_{CC} - I_B R_B - V_{BE} = 0$
(b) $V_{CC} - I_B R_B + V_{BE} = 0$
(c) $V_{CC} + I_B R_B - V_{BE} = 0$
(d) $V_{CC} + I_B R_B + V_{BE} = 10$

31. From Collector-Emitter loop in picture below, what is the Kirchhoff's voltage equation for the loop?



- (a) $V_{CE} + I_C R_C + V_{CC} = 0$
(b) $V_{CE} - I_C R_C + V_{CC} = 0$
(c) $V_{CC} - I_B R_B - V_{BE} = 0$
(d) $V_{CE} + I_C R_C - V_{CC} = 0$

32. Find current I_B of the circuit in figure below?



(a) $16.23\text{ }\mu\text{A}$

(b) $18.23\text{ }\mu\text{A}$

(c) $1.823\text{ }\mu\text{A}$

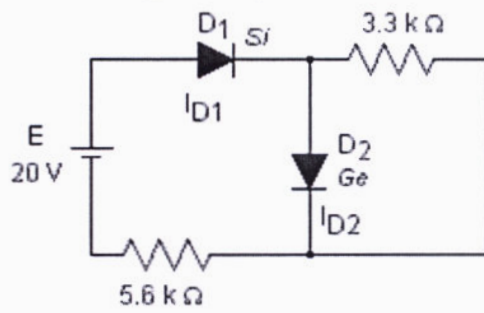
(d) $182.3\text{ }\mu\text{A}$

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Part 2 (25 points) Calculation

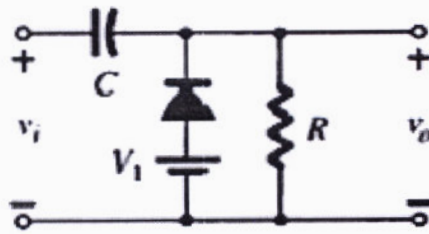
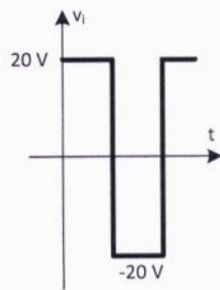
Instruction: Show the mathematical expression and answers of following problems.

1. Determine I_{D1} from figure below

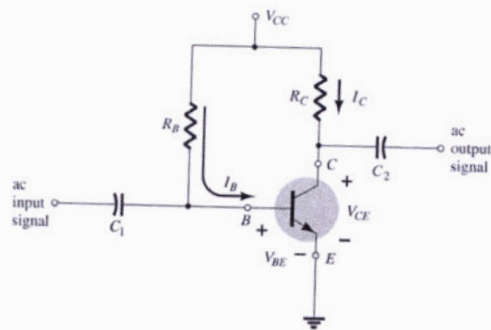
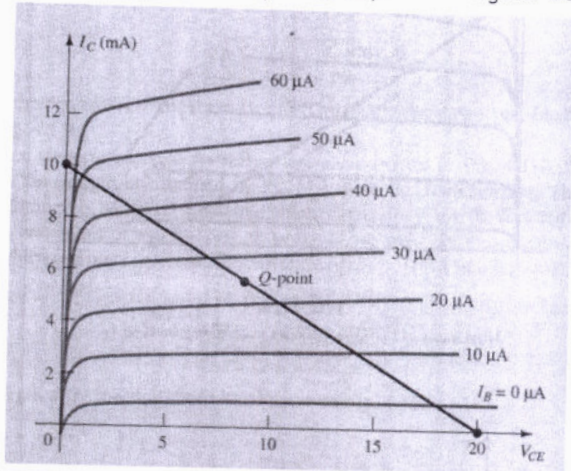


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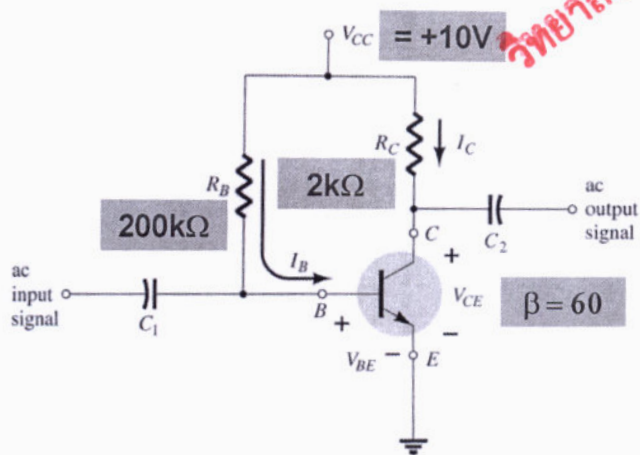
2. Sketch the output waveform of the network below. Given $V_1=5V$. Assume that the diode is ideal.



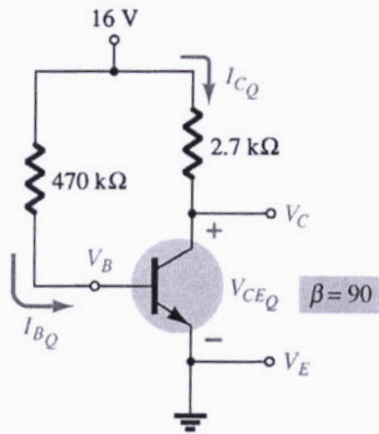
3. Given $I_B = 25\mu\text{A}$ at Q-point in figure below, determine I_C , R_B , R_C of the circuit below.



4. Determine I_B and I_C of the circuit below.



5. Determine $I_{C_{sat}}$ and $V_{CE_{cutoff}}$ of the circuit below.



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Answer Sheet

Name: _____ ID: _____

Subject: 353152 Electronic Circuits I

	a	b	c	d
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